

Zubin Choudhary

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Summary

Senior Product Engineer with 5+ years of experience engineering high-performance systems and developer tooling. Specialized in **Rust (Tauri)** and **TypeScript** ecosystems with a focus on optimization (60fps rendering, memory-safe architecture). Proven track record of migrating legacy infrastructure (Python 2→3 for Fedora) and architecting secure, non-custodial identity protocols. Former **Google Summer of Code** developer with a passion for Open Source and decentralized tech.

Technical Skills

Languages: TypeScript, Rust, Python, Go, Solidity, C++ **Frontend:** React (Hooks/Context), Next.js, UI Virtualization, Canvas API, Tailwind **Backend/Infra:** Node.js, PostgreSQL, Redis, Docker, AWS, REST/GraphQL **Specialized:** Distributed Systems, Applied Cryptography (ERC-4337, EIP-712), Tauri (Desktop), ERP Integration

Professional Experience

FFreed — Founding Engineer

May 2024 – December 2025

- Architected and deployed a developer SDK for automated infrastructure provisioning, reducing integration time for complex authentication flows by **40%**.
- Developed a secure key-sharding recovery system utilizing **Shamir's Secret Sharing (SSS)** and **OAuth**, ensuring high-availability identity management.
- Designed a pluggable **asynchronous relay service**, enabling developers to swap execution providers without modifying core application logic.
- Built a comprehensive developer dashboard and API suite, successfully onboarding 10+ early-stage technical partners for beta evaluation.

Stubs — Senior Software Engineer

December 2022 – October 2023

- Engineered a **high-performance React data grid** capable of rendering **12,000+ dynamic items at 60fps** through custom virtualization techniques.
- Designed and maintained scalable Node.js microservices for processing high-volume transactional data.
- Implemented automated metadata upgrade patterns for digital assets, ensuring data consistency across distributed nodes.
- Developed backend task-queues to handle high-concurrency request spikes during product launches.

Sublime Finance — Software Engineer

December 2021 – November 2022

- Optimized frontend architecture and asset delivery, reducing initial page load times from **8s to 3s (62% improvement)**.
- Integrated complex financial modules with React frontends, focusing on real-time data synchronization and state management.
- Developed secure **off-chain identity verification flows** using EIP-712, balancing user privacy with verifiable data integrity.

Freelance & Open Source — Software Consultant

July 2021 – Present

- Enterprise Tooling:** Developed an internal inventory management and booking system for a regional furniture chain, integrating directly with **Frappe ERP** and reducing manual entry errors by **~90%**.
- System Security:** Implemented End-to-End Encryption (E2EE) for a communication protocol using **Diffie-Hellman** key exchanges over distributed databases (GunDB).
- Infrastructure:** Contributed to cross-chain verification modules for Hyperlane, focusing on replay-protection and security hooks.

Google Summer of Code: Fedora Project — Software Engineering Intern

April 2019 – August 2019

- Led the technical migration of **Goovey Karma** (Fedora's update-testing GUI) from **Python 2 to Python 3**, ensuring long-term maintainability.
- Redesigned the Qt-based user interface to improve developer workflow efficiency for package browsing and bug tracking.
- Packaged and shipped the final tool as an RPM distribution used by Fedora QA and Infrastructure teams.

Key Projects

Holdem — Cross-Platform Desktop Utility (100+ Stars on GitHub)

- Developed a lightweight file-holding utility using **Rust (Tauri)** and **TypeScript**, focusing on low memory footprint and native OS integration.
- Implemented global hotkey listeners and drag-and-drop functionality for seamless file system interaction.

Face-Recognition Public-Transport Wallet

- **Microsoft Imagine Cup Asia Finalist:** Developed a computer vision model (98% accuracy) for automated fare collection using one-shot learning.

Education

B.Tech in Computer Science | Bennett University | *June 2021*